
UNIT 16 SCHEMES FOR AGRICULTURAL DEVELOPMENT

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16.0 OBJECTIVES

After completing this unit, you should be able to:

- describe the need for having agricultural schemes;
- critically assess the agricultural schemes;
- outline the efforts taken by the government to help farmers through the agri based schemes and strategies; and
- explain the impact of the various schemes and programmes.

16.1 INTRODUCTION

The establishment of healthy, sustainable, and inclusive food systems is of utmost importance in order to successfully attain global development

objectives (Fanzo, 2019). According to Dhahri and Omri (2020), the implementation of agricultural development holds significant potential to address extreme poverty, promote inclusive economic growth, and ensure food security for the estimated global population of 9.7 billion individuals by the year 2050. According to Christiaensen *et al.* (2011), the agricultural sector demonstrates a higher efficacy in increasing earnings among the most impoverished individuals, with a growth rate that is two to four times more impactful compared to other sectors. Agriculture has a significant role in fostering economic growth, contributing approximately 4% to the global gross domestic product (GDP). In certain least-developed nations, this sector might even account for more than 25% of the GDP (World Bank, 2023). However, there are concerns over the potential risks to agriculture-driven growth, poverty alleviation, and food security. Various factors, including the repercussions of the COVID-19 pandemic, adverse weather conditions, infestations, and conflicts, are exerting multiple pressures on food systems. Consequently, these pressures are leading to an increase in food costs and exacerbating the issue of hunger. The incursion of Russia into Ukraine has expedited a worldwide food crisis, resulting in the exacerbation of extreme poverty for a significant number of individuals. Approximately 205 million people residing in 45 nations are currently facing severe food scarcity, posing a substantial threat to their well-being and survival.

The potential consequences of climate change on crop yields, particularly in countries with high levels of food insecurity, have been identified in previous studies (McMichael, 2001; Raj et al., 2022). Simultaneously, it is noteworthy that our food systems are accountable for around 30% of total greenhouse gas emissions. The present food systems pose a significant risk to both human health and the environment, as they contribute to excessive pollution and waste, hence undermining sustainability. Approximately 33% of the total world food production is lost or squandered. The mitigation of food loss and waste is of utmost importance in enhancing food and nutrition security as well as in contributing to the achievement of climate objectives and alleviating environmental burdens (McMichael, 2001; Raj et al., 2022).

The primary cause of mortality on a global scale can be attributed to the risks associated with inadequate dietary patterns. A significant portion of the global population faces the challenge of inadequate dietary intake or consumption of nutritionally deficient food, leading to a dual burden of malnutrition. This situation poses a considerable risk of various illnesses and health emergencies. Food insecurity can exacerbate the quality of one's diet and elevate the likelihood of experiencing different types of malnutrition, which may result in both undernutrition and the prevalence of overweight and obesity among individuals. According to Al-Jawaldeh and Abbas (2022), it is approximated that a staggering 3 billion individuals worldwide lack the financial means to access a nutritious meal.

16.2 STATUS OF AGRICULTURE IN INDIA

Agriculture serves as the primary occupation for approximately 58 per cent of India's population. The Gross Value Added (GVA) generated by the sectors of farming, forestry, and fisheries in the fiscal year 2019 (provisional

estimate) is around Rs 18.55 lakh crore (equivalent to US\$ 265.51 billion). The agricultural sector plays a crucial role in facilitating socio-economic development. Agriculture serves as the predominant means of sustenance for around 58 per cent of India's population (IBEF, 2021) with almost 70 per cent of rural households relying only on agricultural activities. The agricultural sector holds considerable importance in the Indian economy, contributing approximately 20% to the Gross Domestic Product (GDP). According to Gupta and Nagar (2017), a significant proportion of India's population, approximately 62 per cent, depends on it for their sustenance. The agricultural industry holds significant importance within the Indian economy, as it provides approximately 20.19 per cent of the Gross Domestic Product (GDP) according to the Annual Report of the Department of Agriculture, Cooperation, and Farmers Welfare for the fiscal year 2020-21. In the early stages of human civilization, agricultural practices mostly served domestic needs. However, as time progressed, advancements in technology and agricultural techniques emerged, leading to increased crop productivity. Consequently, individuals began to derive economic benefits from agricultural pursuits. However, numerous challenges have arisen in the socioeconomic domains, in conjunction with environmental hazards (Bhatt et al., 2019).

The agricultural sector in India has had a gradual drop in its contribution to the country's economy, falling below 15%. This decline can be attributed to the rapid growth rates observed in the industrial and services sectors. However, it is important to note that the significance of the agricultural sector in India extends beyond this particular economic indicator (Dhinakaran & Kesavaran, 2022). Initially, it is worth noting that a significant proportion, approximately 75%, of households in India rely on incomes derived from rural areas. Furthermore, it is worth noting that a significant proportion of India's impoverished population, approximately 770 million individuals or almost 70 per cent, primarily reside in rural regions. Moreover, the food security of India is contingent upon the cultivation of cereal crops, alongside the imperative need to augment the production of fruits, vegetables, and dairy products to cater to the requirements of an expanding populace with escalating incomes. To achieve this objective, an agricultural sector characterized by productivity, competitiveness, diversification, and sustainability must be established rapidly.

India is well recognized as a prominent player in the global agricultural sector. India holds the distinction of being the foremost global producer of milk, pulses, and spices. Additionally, it boasts the largest cow herd, predominantly consisting of buffaloes. Furthermore, India possesses the largest agricultural land dedicated to cultivating wheat, rice, and cotton. The country holds the distinction of being the second most significant contributor to the global production of rice, wheat, cotton, sugarcane, farmed fish, sheep and goat meat, fruit, vegetables, and tea. The nation possesses over 195 million hectares of land dedicated to agricultural cultivation, with rainfed areas accounting for around 63 per cent (nearly 125 million hectares) and irrigated areas constituting 37 per cent (approximately 70 million hectares).

16.3 NEED FOR AGRICULTURAL BASED SCHEMES

India has attained self-sufficiency in food grain production through the implementation of the Green Revolution (Abrol and Sangar, 2006). However, this achievement has resulted in a range of environmental challenges, such as the depletion of soil fertility, waterlogging, pollution of ground and surface water, and the proliferation of pests and diseases (Cummings, 2019). Additionally, the Green Revolution has also led to socioeconomic issues, including the escalation of farm input costs and regional disparities (Cummings, 2019). Furthermore, the issue of climate change has introduced an additional aspect to the preexisting challenges, so presenting a substantial peril to the agricultural sector in India as a whole, and specifically to the nation's food security (Rao et al., 2016). India is recognized as a nation that is particularly susceptible to the impacts of climate change, as indicated by the Indian Network for Climate Change Assessment (INCCA) in 2010. According to a recent body of research conducted by Ray et al. (2019), there has been a notable rise in temperature, an increased occurrence of heat waves, droughts, extreme precipitation events, and intensified cyclonic activities. Saha *et. al.* (2014) have also documented a decline in the intensity of the Indian summer monsoon throughout the later part of the 20th century. This monsoon system serves as a crucial water resource for India's agricultural industry. These modifications have resulted in various implications for the agricultural industry.

Auffhammer et al. (2012) documented a notable decrease in crop productivity in India as a consequence of recurrent droughts occurring since 1966. Additionally, the research findings indicate that the rice output could have experienced an approximate increase of 4 per cent in the absence of elevated nocturnal temperatures and reduced precipitation. According to the research conducted by Guiteras (2009), it was determined that during the short-term period spanning from 2010 to 2039, the impact of climate change on crop yields would result in a decrease ranging from 4.5 to 9 per cent. However, in the long-term period from 2070 to 2099, the absence of adaptation measures would lead to a significant reduction of at least 25 per cent in crop yields. The increase in crop water demand is expected to coincide with extended warming, necessitating an increase in irrigation practices. Nevertheless, the utilization of irrigation and the excessive exploitation of resources have resulted in a significant decline in the groundwater level. This decline persists even in regions that have heightened precipitation as a consequence of climate change. Consequently, if the practice of irrigated agriculture is expanded, it would necessitate the extraction of excessive amounts of groundwater (Zaveri et al., 2016).

The impact of climate change and shifts in agricultural production systems on food security is particularly pronounced for farmers, as highlighted by Soubry et al. (2020). The occurrence of crop loss engenders farmer anguish, inflation, and subsequently, wide-ranging economic ramifications. At present, the annual mean agricultural losses caused just by extreme weather events are leading to estimated losses amounting to approximately 0.25 per

cent of India's Gross Domestic Product (GDP) (Singh et al., 2019). Moreover, it has been contended that climatic stressors have the potential to intensify the financial obligations of farmers, leading to an increased load of debt, and consequently, prompting some individuals to resort to suicide as a coping mechanism (Carleton, 2017). Furthermore, it is worth noting that there was a significant decrease in the number of individuals engaged in cultivation activities throughout India. This loss was so sudden and pronounced that for the first time since the country gained independence, the number of agricultural labourers exceeded that of farmers (Gupta, 2016). Despite experiencing a decrease in its contribution to the nation's Gross Domestic Product (GDP), decreasing number of individuals engaged in cultivation and instances of farmers' suicide, it is noteworthy that the climate-sensitive agricultural sector remains the predominant means of sustenance in India.

Given the aforementioned factors, it is crucial to acknowledge the susceptibility of Indian agricultural producers to a range of risks, particularly the threat posed by climate change. In light of these circumstances, it becomes imperative to establish comprehensive programmes that can effectively assist farmers in mitigating and overcoming these challenges.

16.4 IMPORTANCE OF AGRI BASED SCHEMES AND STRATEGIES

Historically, the developmental approach towards the agriculture sector in India has predominantly centred on augmenting agricultural production and enhancing food security. India has had a notable outcome, characterized by a 45 per cent rise in per capita food output. This achievement has not only ensured India's self-sufficiency in food production as a whole but has also positioned the country as a net exporter of food.

The emergence of agrarian distress in the country throughout the 1990s, which reached a significant level in certain years, can be attributed to the low absolute income levels and the substantial and worsening income discrepancy between farmers and non-agricultural workers. The nation also experienced a significant rise in the occurrence of suicides among farmers from 1995 to 2004. This distressing trend has been attributed to various variables, including financial losses in agricultural activities, sudden fluctuations in farm revenue, and persistently low levels of income derived from farming. The adverse impact of volatile and insufficient farm revenue is leading to a decline in interest in agricultural activities and investments, thereby compelling a growing number of farmers, especially those in younger age brackets, to abandon the profession. The aforementioned phenomenon has the potential to significantly impact the future of agriculture in the nation. The significance of agricultural revenue for addressing rural distress and promoting farmers' well-being is evident (Chand, 2016). Within this context, the government has implemented a range of initiatives and policies aimed at promoting the well-being of farmers.

16.5 AGRICULTURE BASED SCHEMES

The agricultural sector in India has experienced significant economic policy adjustments since gaining independence. The changes and subsequent policy interventions primarily focus on many aspects such as the growth rate of agricultural output, ensuring food security, improving nutrition, promoting regional equity, maintaining price stability, enhancing farm income, and evaluating the impact on trade. The Government of India is placing significant emphasis on the welfare of farmers and is actively implementing various initiatives aimed at revitalizing the agricultural sector and enhancing the economic well-being of farmers.

Check Your Progress Exercise 1

Note: a) Use the spaces given below for your answers.

b) Check your answer with those given at the end of the unit.

1. Write a brief note on agricultural status in India.

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2. Briefly describe the need for having agri-based schemes.

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3. Briefly describe about importance of agri-based schemes and strategies.

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16.6 VARIOUS PROGRAMS AND SCHEMES IN AGRICULTURAL SECTOR IN INDIA

The government implements a number of developmental programs, schemes, reforms, and policies for the benefit of farmers, such as modernizing and rationalizing input use to reduce costs, increasing crop production, remunerative returns, and income support, all of which have been able to augment farmers' income directly or indirectly. The following is a list of major agricultural schemes and programs in India.

16.6.1 Doubling Farmers Income

The governmental initiatives aimed at doubling the income of farmers have demonstrated significant outcomes. The Government established an Inter-Ministerial Committee in April 2016 to investigate matters pertaining to the "Doubling of Farmers Income (DFI)" and propose measures for its attainment. In September 2018, the Committee presented a comprehensive report consisting of 14 volumes to the Government. This study included a strategic approach aimed at achieving a twofold increase in farmers' income. The proposed plan encompassed a range of policies, reforms, and programs. In order to accomplish the stated target, the Committee has identified seven areas of income development.

Aims of DFI

- An enhancement in crop productivity has been observed.
- An improvement in livestock productivity has been noted. • There has been an increase in resource use efficiency, leading to a reduction in production costs.
- An increase in cropping intensity has been observed. There has been a shift towards engaging in high-value agriculture for diversification purposes. Farmers have been able to obtain remunerative prices for their produce. There has been a migration of surplus manpower from farm occupations to non-farm occupations.

The premise of the strategy for doubling farmers' income is based on the following primary principles:

- Enhancing overall agricultural output by achieving greater productivity
- Streamlining and minimizing production costs
- Ensuring profitable prices for agricultural products
- Implementing effective risk management strategies
- Embracing sustainable technology

Roadmap and action plan

The quantitative framework utilized for the purpose of doubling farmers' income has successfully identified a total of seven distinct sources of growth. The following items are:

- Rise in the productivity of crops.
- Increase in the quantity of livestock produced.
- The efficiency of input utilization has improved, resulting in cost savings for farmers.
- Increase in crop intensity, indicating a higher frequency of crop cultivation within a given area.
- Diversifying the agricultural activities towards the cultivation of high value crops.
- Improved price realization for their agricultural produce.

- A trend has been seen in the shift of cultivators from traditional farming occupations to non-farm jobs.

16.6.2 Soil Health Card Scheme

Introduced in 2015, the scheme was implemented with the aim of aiding State Governments in the issuance of Soil Health Cards to farmers nationwide. The Soil Health Cards offer farmers valuable information regarding the nutrient state of their soil, as well as recommendations for the optimal dosage of fertilizers to enhance soil health and fertility. The purpose of a Soil Health Card is to evaluate the present condition of soil health and, when employed longitudinally, to ascertain alterations in soil health that are influenced by land management practices. The Soil Health Card is a comprehensive tool that presents many soil health indicators along with their corresponding descriptive phrases. The indicators are commonly derived from the empirical knowledge and expertise of farmers, who possess a deep understanding of the local natural resources. The card enumerates soil health indicators that can be evaluated without the assistance of specialized technical or laboratory equipment.

Soil Health Card (SHC) is a Government of India scheme promoted by the Department of Agriculture & Co-operation under the Ministry of Agriculture and Farmers' Welfare. It is being implemented through the Department of Agriculture of all the State and Union Territory Governments.

The Soil Health Card Scheme is an advantageous initiative that offers significant benefits to farmers. India is home to a significant number of farmers. Furthermore, there is a lack of knowledge regarding the optimal selection of crop varieties that would result in the highest possible yield. The individuals lack knowledge of the characteristics and composition of their soil. Individuals may possess knowledge based on personal experience regarding which crops are successful and which crops are unsuccessful. However, their knowledge of potential strategies to enhance soil quality remains limited.

Benefits of the scheme:

- The program systematically observes the soil conditions of the farmers and provides them with a standardized report. Thus, individuals are able to make informed decisions on the selection of crops to cultivate and those to forgo.
- The soil is subject to frequent monitoring by the authorities. A report is sent to farmers on a triennial basis. Farmers need not be concerned if the composition of the soil undergoes alterations as a result of several circumstances. Furthermore, they will consistently possess up-to-date information on the condition of their soil.
- The responsibilities of the government extend beyond merely compiling a list of measures necessary for enhancing soil quality. Furthermore, it is worth noting that they also engage the services of knowledgeable professionals to assist farmers in implementing the necessary remedial actions.

- The Soil Health Card Scheme is expected to provide farmers with an accurate record of soil health. Additionally, individuals have the opportunity to engage in the study of soil management strategies. Consequently, individuals are able to strategize for the future of their agricultural produce and land.
- Typically, under governmental initiatives, there is a tendency for the individual responsible for conducting a study on behalf of a specific farmer to undergo changes. However, within the framework of the Soil Health Card Scheme, the government is emphasizing the importance of ensuring consistency in the individual responsible for doing soil analysis for each farmer. This will additionally augment the efficacy of the program.
- The soil analysis report will provide farmers with a comprehensive understanding of the specific nutrients that are deficient in their soil. Consequently, determine the crops to which they should allocate their investments. Additionally, they will provide information regarding the specific fertilizers required. Ultimately, it is anticipated that there will be an increase in crop yield.
- The primary objective of the system is to ascertain the specific soil type. Subsequently, it is vital to present strategies that might be employed to enhance the aforementioned situation. Despite the presence of certain constraints, some measures may be taken to maximize the productivity of a given soil. This strategy is being implemented by the government in an effort to achieve its objectives.

16.6.3 Pradhan Mantri Fasal Bima Yojana (PMFBY)

The Pradhan Mantri Fasal Bima Yojana (PMFBY) was initiated in the year 2016. The scheme in question offers protection against financial losses incurred by farmers as a result of a range of adverse occurrences. This encompasses the occurrence of crop failure resulting from localized hazards, post-harvest losses, natural disasters, unanticipated precipitation patterns, pests, crop ailments, and similar factors. The program functions with the guiding principle of "One Nation, One Crop, and One Premium" and endeavours to offer economically viable crop insurance options to farmers throughout India. The objective of the Pradhan Mantri Fasal Bima Yojana is to enhance the reach of crop insurance in India, with a primary emphasis on providing coverage for the whole cultivated land in the nation.

Objective of Pradhan Mantri Fasal Bima Yojana

The primary objective of the Pradhan Mantri Fasal Bima Yojana (PMFBY) is to provide assistance and promote the practice of sustainable agricultural output within the agriculture sector. The mitigation of financial losses resulting from crop failure contributes to the stabilization of farmer incomes. When farmers achieve a consistent income, they are able to allocate resources towards enhancing their agricultural methods, resulting in several advantageous outcomes.

Coverage of PMFBY crop insurance scheme

Encompasses stages of crop cultivation as well as the associated risks that may result in crop loss.

Basic Product: The scheme provides coverage for the risk associated with the standing crop from the time of sowing through harvesting. Comprehensive risk insurance is offered to mitigate yield losses on an area-based strategy, resulting from uncontrollable risks such as drought, dry spells, flood, inundation, widespread pests and diseases, landslides, fire caused by natural factors, lightning, storms, hailstorms, and cyclones.

Add-On Coverage: In addition to the obligatory fundamental product, the state governments and union territories, in collaboration with the State Level Coordination Committee on Crop Insurance (SLCCCI), have the option to select any or all of the supplementary coverage options listed below. These options are chosen based on the specific crop and geographical requirements of their respective states, in order to mitigate risks associated with various stages of crop growth and potential crop loss.

Prevented Sowing/ Planting Risk: If over 75% of a designated crop area within an insurance unit remains unplanted, or if there is a failure of germination during the sowing period as a result of bad weather conditions such as deficient rainfall or unfavourable seasonal circumstances, farmers are entitled to mitigate the risk of sowing failure. The lump sum payment provided to insured farmers through this coverage would amount to 25% of the insured total, resulting in the termination of the insurance policy.

Mid-Season Adversity: Losses may occur in the event of unfavourable seasonal conditions, such as floods, extended periods of dry weather, and severe drought. These factors might result in a crop yield that is predicted to be less than 50% of the normal production for the season. The indemnification provided to farmers who are insured under this policy would amount to 25% of the anticipated claims, with the possibility of being adjusted in accordance with the final claims.

Post-Harvest Losses: The insurance coverage is limited to a maximum duration of two weeks (14 days) following the harvest for crops that necessitate drying in a cut and spread or tiny packaged state on the field. This coverage protects against specific risks such as hailstorms, cyclones, cyclonic rains, and unseasonal rains. The evaluation of damage will be conducted on a per-farm basis.

Localized Calamities: The loss or damage to insured crops is caused by specific localized risks such as hailstorms, landslides, flooding, cloud bursts, and natural fires caused by lightning. These risks mostly impact isolated farms within the designated area.

Table 16.1: State wise Crop Insurance Statistics

FY 2019-20 - PMFBY & RWBCIS Combined - State Wise Business Statistics as on 31.08.2021								
State/UT Name	Farmers Application s Insured (Lakh)	Area Insured (Lakh ha)	Sum Insured	Farmers Share in Premium	Gross Premium	Reported Claims	Paid Claims	Farmer Application s Benefitted (Lakh)
Rs. Crore								
A & N Islands	0.001	0.001	0.32	0.002	0.03	0.00	-	-
Andhra Pradesh	27.884	19.873	15,031.92	0.172	1,474.73	1,259.01	1,254.03	13.533
Assam	10.027	5.615	4,033.73	57.099	142.44	17.27	-	-
Bihar	-	-	-	-	-	-	-	-
Chhattisgarh	40.177	24.346	9,032.42	180.861	1,245.79	1,314.60	1,296.59	15.025
Goa	0.009	0.001	0.96	0.022	0.04	0.01	0.01	0.001
Gujarat	24.810	29.438	16,143.17	467.959	3,614.98	354.89	111.67	0.927
Haryana	17.111	22.505	15,132.97	268.803	1,221.72	932.26	927.45	5.552
Himachal Pradesh	2.840	0.941	746.68	30.711	83.07	64.60	58.01	1.505
Jammu & Kashmir	-	-	-	-	-	-	-	-
Jharkhand	10.921	6.451	3,739.05	2.792	356.02	25.46	-	-
Karnataka	21.316	21.668	9,830.01	253.480	2,276.33	1,357.79	1,215.35	6.869
Kerala	0.581	0.372	307.80	6.117	72.50	85.90	85.90	0.457
Madhya Pradesh	78.929	112.682	32,030.72	629.279	3,758.65	5,905.48	5,811.75	30.546
Maharashtra	145.642	79.223	30,282.21	868.432	6,353.77	6,755.92	6,747.05	87.895
Manipur	0.033	0.026	17.34	0.347	1.26	1.14	1.14	0.032
Meghalaya	0.006	0.003	2.31	0.086	0.09	0.18	0.18	0.005
Odisha	48.769	18.688	12,197.35	241.876	2,132.85	1,177.91	1,139.48	12.078
Puducherry	0.120	0.092	62.30	-	4.18	7.16	-	-
Rajasthan	85.260	96.935	34,909.48	734.677	5,060.10	4,920.44	4,920.31	25.574
Sikkim	0.000	0.000	0.08	0.002	0.00	-	-	-
Tamil Nadu	38.705	14.072	9,329.88	168.419	1,923.39	1,090.13	1,056.84	13.213
Telangana	10.335	11.347	8,459.17	239.487	880.75	402.28	-	-
Tripura	0.364	0.061	37.17	0.756	1.07	0.81	0.80	0.077
Uttar Pradesh	46.947	35.572	16,743.95	321.955	1,304.82	1,116.75	1,092.74	9.343
Uttarakhand	2.127	1.135	968.53	28.208	113.71	103.18	103.17	0.949
West Bengal	-	-	-	-	-	-	-	-
GRAND TOTAL	612.9	501.0	219,040	4,502	32,022	26,893	25,822	223.6
* 2019-20 claims yet to be fully reported								
# Information is based on declarations received from Implementing Insurance Companies and is currently being verified by Department of Agriculture, Cooperation and Farmers Welfare, Government of India with concerned State Governments								
\$ Majority of claim settlement is pending due to pending State subsidy and/or pending yield data. Some claims are also pending due to issues such as payment failure, discrepancies in yield data etc.								

Source: <https://pmfby.gov.in/pdf/2021-08-31%20-%202019-20.pdf>

Pradhan Mantri Fasal Bima Yojana Premium Charges

The Pradhan Mantri Fasal Bima Yojana Premium entails an equal sharing of the premium payment between the farmer and the government. The financial responsibility of farmers towards the premium for the Pradhan Mantri Fasal Bima Yojana (PMFBY) crop insurance is determined based on the above table. The remaining portion of the premium is covered by the State and Central Government.

Sr.	Cropping Season	Type of Crop	Max Premium Payable By Farmer (% Of Sum Insured)
1	Kharif	All food grain and Oilseed crops (all Cereals, Millets, Pulses and Oilseed crops)	2.0% of SI or Actuarial rate, whichever is less
2	Rabi	All food grain and Oilseed crops (all Cereals, Millets, Pulses and Oilseed crops)	1.5% of SI or Actuarial rate, whichever is less
3	Kharif & Rabi	Annual Commercial/Horticultural crops	5% of SI or Actuarial rate, whichever is less

16.6.4 Weather Based Crop Insurance Scheme (WBCIS)

The Weather Based Crop Insurance Scheme (WBCIS) aims to offer insurance coverage to farmers against unfavourable weather conditions, including insufficient or excessive rainfall, extreme temperatures, humidity, and other factors that are known to have a negative impact on crop yields. One advantage of this approach is its ability to expedite the resolution of disputes in a timely manner.

Objectives and benefits

- a) This proposal entails the provision of insurance coverage for designated food crops, oilseeds, and annual horticultural/commercial crops. Additionally, it suggests the implementation of a standardized maximum premium rate applicable to all farmers, similar to the Pradhan Mantri Fasal Bima Yojana (PMFBY).
- b) Kharif season - 2% of the sum insured.
- c) Rabi Season 1.5% of the sum insured.
- d) Commercial/horticultural crops 5% of the sum insured.

Crops covered

- Major Food crops (Cereals, Millets & Pulses) & Oilseeds, Commercial / Horticultural crops

Farmers Covered

All farmers including sharecroppers and tenant farmers growing the notified crops in the notified areas are eligible for coverage. However, farmers should have an insurable interest in the insured crop. The non-loanee farmers are required to submit necessary documentary evidence of land records and/or applicable contract/agreement details (in the case of sharecroppers/tenant farmers). All farmers availing Seasonal Agricultural Operations (SAO) loans from Financial Institutions (i.e., loanee farmers) for the crop(s) notified are covered on a compulsory basis. The scheme is optional for non-loanee farmers.

16.6.5 National Mission for Sustainable Agriculture (NMSA)

The maintenance of agricultural output is contingent upon the quality and accessibility of natural resources, such as soil and water. The promotion of conservation and sustainable utilization of limited natural resources through context-specific approaches can contribute to the long-term sustainability of agricultural growth. The agricultural sector in India continues to be mostly dependent on rainfall, including over 60% of the nation's cultivated land and contributing to 40% of the overall food output. Therefore, the preservation of natural resources, together with the advancement of rainfed agriculture, is crucial in order to address the increasing need for food grain inside the nation. In order to achieve this objective, the National Mission for Sustainable Agriculture (NMSA) has been developed with the aim of improving agricultural productivity, particularly in rainfed regions. The mission places emphasis on integrated farming, water usage efficiency, soil

health management, and the harmonization of resource conservation efforts.

The primary objective of implementing this method is to augment agricultural productivity, namely within the domain of pastureland. The primary emphasis is directed towards the application of mixed farming techniques, the efficient usage of water resources, the adoption of soil quality management strategies, and the conservation of natural resources. The objective of this paper is to examine a range of significant principles in the field of agriculture. These principles encompass the effective utilization of water, management of nutrition, promotion of diversified livelihoods through the adoption of environmentally sustainable and energy-efficient equipment, conservation of natural resources and the implementation of mixed farming practices.

Schemes under NMSA

RAD - Rainfed Area Development, SHM - Soil Health Management, SMAF - Sub Mission on Agro-Forestry, PKVY – Paramparagat Krishi Vikas Yojana, SLUSI - Soil and Land Use Survey of India, NRAA - National Rainfed Area Authority, MOVCDNER - Mission Organic Value Chain Development in North Eastern Region, NCOF - National Centre of Organic Farming.

16.6.6 Pradhan Mantri Krishi Sinchai Yojana

The formulation of the PMKSY initiative involves the amalgamation of pre-existing schemes, specifically the Accelerated Irrigation Benefit Programme (AIBP) administered by the Ministry of Water Resources, River Development & Ganga Rejuvenation, the Integrated Watershed Management Programme (IWMP) overseen by the Department of Land Resources, and the On Farm Water Management program under the Department of Agriculture and Cooperation. The execution of the project will be undertaken by the Ministries of Agriculture, Water Resources, and Rural Development. The fundamental objective of the Ministry of Rural Development is around the implementation of many initiatives, including rainwater conservation, farm pond construction, water harvesting structure establishment, minor check dam creation, and contour bunding measures.

The Ministry of Water Resources, River Development, and Ganga Rejuvenation (MoWR, RD & GR) is planning to execute a series of projects to build dependable irrigation sources. These activities encompass the construction of diversion canals, field channels, and water diversion/lift irrigation systems, alongside the installation of water distribution networks. The Ministry of Agriculture aims to promote the implementation of efficient water transportation systems and precision water application equipment including drip irrigation, sprinklers, pivots, and rain guns, in the agricultural industry. The endeavour, referred to as "Jal Sinchan," encompasses the establishment of micro-irrigation infrastructure to supplement endeavors aimed at generating water resources. Furthermore, the ministry seeks to engage in extension operations in order to facilitate the adoption of moisture conservation techniques and agronomic measures that are grounded in scientific principles.

The main aim of the Pradhan Mantri Krishi Sinchayee Yojana (PMKSY) is to enhance the availability of irrigation facilities to agricultural fields throughout the country. The primary objective of this program is to optimize agricultural output through the promotion of effective water resource management. By doing so, it hopes to empower farmers to attain increased crop yields in relation to the amount of water utilized. As a result, this would lead to the anticipated improvement in the economic well-being of rural areas. The Prime Minister Krishi Sinchayee Yojana (PMKSY) program has been formulated to give priority to the effective implementation of a holistic solution for the irrigation supply chain. The proposed approach has multiple elements, including water supplies, distribution networks, effective farm-level applications, extension services on new technologies, and information dissemination. The development of these components is predicated upon a comprehensive planning process that is undertaken at both the district and state levels.

Objectives of the scheme

- Use proper agricultural practices to conserve soil and water.
- Collect rainwater by building check dams and ponds.
- In water-logged locations, practice crop diversification, seed production, and nursery raising.
- Use drip and sprinkler irrigation systems to save 30-37% on water while improving crop quality and output.

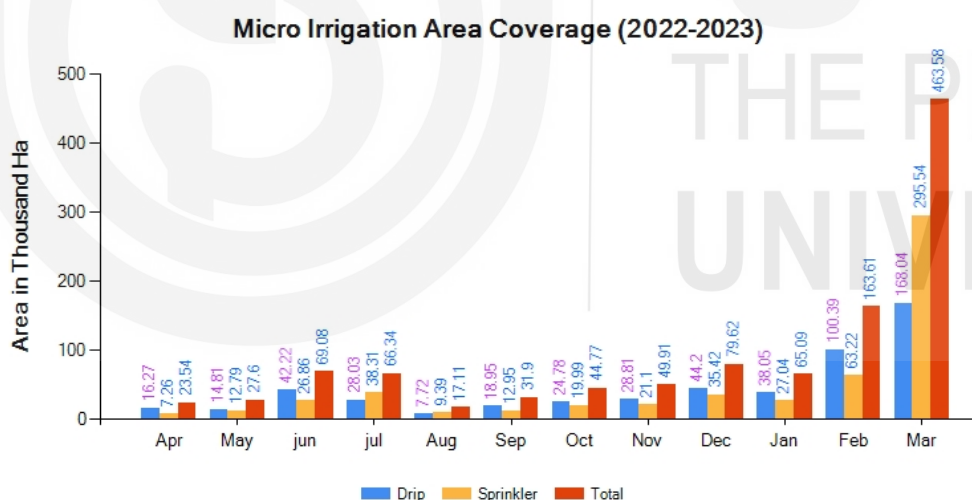


Fig. 16.1: Micro irrigation area coverage in 2022-23

(Source: <https://pmkasy.gov.in/AboutPMKSY.aspx>)

16.6.7 Kisan Credit Card (KCC) Loan Scheme

According to the most recent revision by the Union Finance Minister, a total of 1.82 crore credit cards were sanctioned to farmers under the credit card lending scheme until February 29th, 2020. Credit The Kisan financial Card scheme was initiated by the Indian government in 1998 to facilitate prompt financial access to farmers to fulfil their agricultural needs. These credit cards are associated with the Pradhan Mantri Kisan Samman Nidhi Yojana to facilitate the beneficiaries in accessing the benefits of the credit card conveniently.

The scheme in question facilitates farmers in accessing short-term formal credit and was implemented by NABARD (National Bank for Agriculture and Rural Development). Under this scheme, farmers are able to secure a loan of up to 300,000 units at an interest rate of 4%. The Government of India provides credit notes for not only agricultural activities but also for animal husbandry and fisheries with the aim of addressing the diverse expenses associated with conducting these businesses. Typically, financial institutions extend loans to agriculturalists at a somewhat elevated rate of interest. However, via the implementation of the Kisan credit card program, these banks offer loans at a reduced interest rate, with a minimum of 2% and an average of 4%.

16.6.8 Paramparagat Krishi Vikas Yojana (PKVY)

The Paramparagat Krishi Vikas Yojana (PKVY) was introduced in 2015 as an expansion of the Soil Health Management (SHM) initiative inside the Centrally Sponsored Scheme (CSS), known as the National Mission on Sustainable Agriculture (NMSA). The primary objective of the Paramparagat Krishi Vikas Yojana (PKVY) is to provide assistance and encouragement for the practice of organic farming, with the ultimate goal of enhancing soil health.

The initiative advocates for the implementation of the Participatory Guarantee System (PGS) for India (PGS-India), a method of certifying organic products that emphasizes trust, local relevance, and the active participation of producers and consumers in the certification process. PGS-India operates independently from the "Third Party Certification" structure.

The funding allocation under the scheme follows a ratio of 60:40, with the Central Government contributing 60% and the State Governments contributing 40%. In the context of North Eastern and Himalayan States, central help is allocated in a ratio of 90:10 (Centre: State), while for Union Territories, the help is granted at a rate of 100%.

Approach

The implementation of PKVY adheres to the recommended conversion period of 36 months from conventional farming to organic farming, as outlined by PGS-India. This conversion phase spans a time frame of three years.

PKVY, or the Paramparagat Krishi Vikas Yojana, effectively facilitates the adoption of PGS certification by employing a clustering method to advance organic farming practices. PGS certification enables farmers to obtain certification for their organic produce, allowing them to appropriately label and promote their products within the local market.

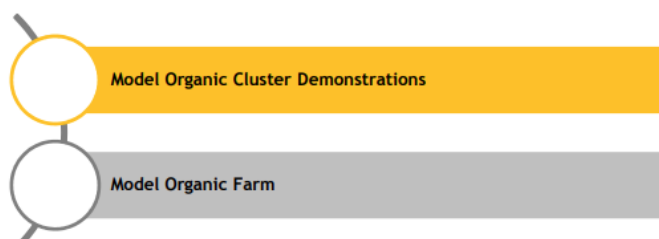


Fig. 16.2: Key Components

16.6.9 Mission Organic Value Chain Development for North Eastern Region

The Ministry of Agriculture and Farmer Welfare has initiated a Central Sector Scheme called "Mission Organic Value Chain Development for North Eastern Region" to tap into the potential of organic farming in the states of Arunachal Pradesh, Assam, Manipur, Meghalaya, Mizoram, Nagaland, Sikkim, and Tripura during the 12th plan period. The objective of the scheme is to facilitate the growth of certified organic production through a value chain approach, which connects producers with consumers. Additionally, it seeks to foster the development of the entire value chain encompassing various stages such as inputs, seeds, and certification, as well as the establishment of facilities for collection, aggregation, processing, marketing, and brand-building efforts.

Mission Goals

- The objective is to establish a minimum of one to two reproducible end-to-end organic value chain models in each state by incorporating producers, handlers, processors, and market facilitation organizations.
- To enhance the capabilities of 30,000 to 50,000 farmers in the northeastern region by establishing around 100 farmer-producer firms. These companies will be provided with complete control of the value chain.
- Transition from subsistence farming to commercial organic farming by implementing comprehensive facilities throughout the entire process.
- To establish the Northeastern states as significant contributors of organic commodities for both domestic and global markets.

Salient Features

- Recognizing the potential of organic farming in the North Eastern Region of the country.
- The Department actively advocates for the adoption and implementation of organic farming practices across the North Eastern States.
- The North Eastern States of India is poised to emerge as a prominent organic centre within the country.
- The Farmers Interest Group (FIG) or Farmers Producer Companies (FPCs) will be provided with the necessary resources and infrastructure for the collection, aggregation, post-harvest processing, and market integration of agricultural products.
- A total of 50,000 farmers are expected to experience positive outcomes as a result of their involvement in various aspects of the value chain, including production, processing, marketing, and support services provided by value chain agencies.
- To facilitate the development of a unique brand of organic produce within the NE Region.
- The region boasts a remarkably diverse array of biological species, offering a valuable opportunity to study a wide range of crop alternatives.

16.6.10 National Agriculture Market (e-NAM)

The National Agriculture Market (eNAM) is an electronic trading system that operates across India, connecting the current Agricultural Produce Market Committee (APMC) mandis to establish a cohesive national market for agricultural commodities. The Small Farmers Agribusiness Consortium (SFAC) serves as the primary organization responsible for the implementation of the National Agriculture Market (eNAM) program, operating under the authority of the Ministry of Agriculture and Farmers' Welfare within the Government of India.

Further details on e-NAM have been comprehensively provided in Unit 4.

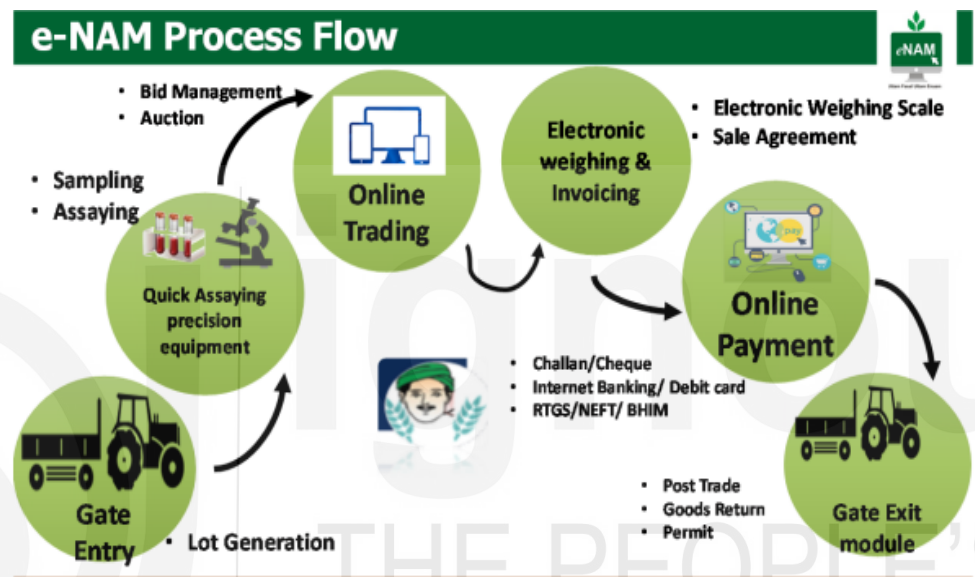


Fig. 16.3: e-NAM process flow (Source: e-NAM)

16.6.11 Pradhan Mantri Kisan Maan Dhan Yojana (PM-KMY)

The Government of India has introduced the Pradhan Mantri Kisan Maan Dhan Yojana (PM-KMY), a pension system aimed at offering social security to small and marginal farmers. This program is specifically designed for those engaged in agricultural activities within the age range of 18 to 40 years. Its primary objective is to establish a sustainable revenue stream for farmers even during periods when they are unable to actively engage in fieldwork. The initiative was introduced in September 2019 as a component of the government's endeavours to mitigate the welfare and financial requirements of agricultural workers.

The scheme is under the regulatory oversight of the Department of Agriculture. The farmer is required to enrol in the pension fund. To derive advantages from this particular system, it is necessary to engage in certain actions or follow specific guidelines. This program presents a novel initiative for agricultural practitioners, providing them with a guaranteed monthly salary of INR 3,000 at reaching the age of 60.

16.7 IMPACTS OF AGRICULTURAL SCHEMES

Given that agriculture falls under the purview of the State, the aforementioned plans and programs must be executed collaboratively with the respective State Governments. The government facilitates these initiatives through increased budgetary allocation and the provision of non-budgetary financial resources, such as the establishment of Corpus Funds like the Micro Irrigation Fund, Agriculture Infrastructure Funds, PM Matasya Sampada Yojana, Animal Husbandry Infrastructure Development Fund, as well as the formation of Farmer Producer Organizations (FPOs) and Grameen Agriculture Markets. Moreover, as a result of the execution of these programmes, there has been a significant increase in the production of food grains as well as horticultural crops. Additionally, the nation has experienced a significant increase in the export of agricultural and related commodities. These achievements have been made possible as a result of a significant increase in budget allocation for the agricultural sector, rising from Rs. 27662.67 crores in the fiscal year 2013-14 to Rs. 1,32,513.62 crores in the fiscal year 2022-23. Based on the most recent Economic Survey, it has been observed that the agricultural and allied industries saw a favourable growth rate of 3.6 per cent in the fiscal year 2020-21.

In relation to the aforementioned matter, it has been indicated that the Ministry of Statistics and Programme Implementation, specifically the National Statistical Office (NSO) carried out a Situation Assessment Survey (SAS) of agricultural households during the 70th round of the National Sample Survey (NSS) from January 2013 to December 2013. This survey was conducted concerning the agricultural year from July 2012 to June 2013. Additionally, a similar survey was conducted during the 77th round of the NSS from January 2019 to December 2019, with reference to the agricultural year from July 2018 to June 2019. These surveys were conducted exclusively in the rural areas of the country. The projected average monthly income per agricultural household was determined based on surveys conducted during the NSS 70th round (2012-13) and NSS 77th round (2018-19), yielding figures of Rs. 6,426/- and Rs. 10,218/- correspondingly.

The government's endeavours to effectively administer these plans have thus far had favourable outcomes, leading to an improvement in farmers' income. As a component of the 'Azadi ka Amrit Mahotsav' initiative, the Indian Council of Agricultural Research (ICAR) has published a book that comprises a compilation of success narratives from a subset of 75,000 farmers who have experienced a significant boost in their earnings, surpassing a twofold growth amidst a larger population of prosperous farmers.

16.8 ANALYSIS OF VARIOUS SCHEMES AND PROGRAMS

Advantages

- The implementation of subsidies for agricultural infrastructure, minimum support prices (MSP), and bank lending is expected to have a significant

impact on farmers' income.

- The minimum support price (MSP) plays a crucial role in the agricultural industry as it helps to mitigate costs and ensures the attainment of food security.
- The implementation of enhanced irrigation systems to foster sustainability and save water resources.
- Government initiatives and economic measures aimed at enhancing income levels, stimulating economic growth, fostering job creation, and increasing agricultural output.
- The impact of the agriculture program underscores the significance of previous welfare initiatives, rural infrastructure development, and financial inclusion in order to attain accelerated socio-economic progress.

Benefits

- The primary advantage of these agricultural schemes is in the farmers' ability to readily access loans and subsidies for their production needs promptly. The implementation of Kisan sprinkler irrigation and drip irrigation systems has the potential to enhance the efficient utilization of water resources while simultaneously preserving natural resources.
- Under the E-NAM system, farmers have the opportunity to engage in online trading of their agricultural products directly with customers, eliminating the need for intermediaries and associated brokerage fees.
- Enhancing productivity is expected to have a positive impact on the economic growth and gross domestic product (GDP) of the nation.
- The enhancement of agricultural infrastructure and the provision of loans for innovative agricultural machinery has the potential to increase productivity while reducing the need for labour, thereby benefiting farmers in the nation.

Constraints

- There is a necessity to foster knowledge of agricultural literacy among farmers, as well as to disseminate information about the diverse schemes and their associated benefits.
- The implementation of agricultural projects by the government should be accompanied by stringent procedures to ensure that the intended beneficiaries can derive maximum benefits.
- The regulations and protocols pertaining to government-issued agricultural credit should be relaxed, allowing for greater accessibility. Additionally, a reduction in the interest rates imposed on these credit facilities is warranted.

Check Your Progress Exercise 2

Note: a) Use the spaces given below for your answers.

b) Check your answer with those given at the end of the unit.

1. Describe various Agri-based schemes floated by the Government of India.

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2. How do various Agri-based schemes help the farmers in India?

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3. What are the various challenges pertaining to agri-based schemes in India?

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16.9 LET US SUM UP

Agriculture serves as the fundamental foundation for the provision, generation, processing, marketing, and dissemination of food resources. Agricultural commodities have a significant role in bolstering the Gross Domestic Product (GDP) and fostering employment opportunities within rural regions. The advent of modern agricultural practices has had a transformative impact on the livelihoods of farmers. The Indian government has implemented a range of policies and initiatives, including the Minimum Support Price (MSP), Market Intervention Fund (MIF), Pradhan Mantri Krishi Sinchai Yojana (PMKSY), Pradhan Mantri Fasal Bima Yojana (PMFBY), electronic National Agriculture Market (e-NAM), Pradhan Mantri Kisan Samman Nidhi (PM-KISAN), Pradhan Mantri Jan Dhan Yojana (PMJDY), Pradhan Mantri Kisan Urja Suraksha evam Utthaan Mahabhiyan (PM-KUSUM), Paramparagat Krishi Vikas Yojana (PKVY), National Agriculture Market Scheme (NAMS), and Mahatma Gandhi National Rural Employment Guarantee Scheme (MGNREGS).

The utilization of the mobile application Kisan Suvidha, together with the implementation of innovative initiatives such as Kisan Rail and Krishi Udaan, contribute to the objective of doubling farmers' income (DFI). These interventions contribute to the enhancement of rural economies by promoting

the expansion of irrigation coverage, facilitating access to crop insurance, and fostering income stability. Additionally, they are responsible for ensuring the provision of financial assistance, facilitating the distribution of credit, and implementing the Direct Benefit Transfer system to allocate subsidies and money to eligible recipients. The utilization of contemporary technology in conjunction with farm-based practices such as poultry, dairy, forestry, and beekeeping, together with the assistance provided by Self-Help Groups (SHGs), is expected to have a direct influence on the productivity, profitability, financial inclusion, and overall well-being of farmers in the 21st century. Furthermore, this approach is anticipated to contribute significantly to the economic development of the country.

16.10 KEYWORDS

Alleviating	: Reduction in size; like poverty reduction.
De-Centralized	: Distribution of power equitably.
Direct Benefit Transfer	: The primary objective of this Direct Benefit Transfer program is to increase transparency and eliminate embezzlement in the distribution of Central Government of India-sponsored funds. Below-poverty-line citizens will receive direct benefits or subsidy transfers through DBT.
Eradicating	: Elimination of something; like poverty and malnutrition.
Farmers Interest Group (FIG)	: A Farmer Interest Group (FIG) refers to a collective of farmers who operate autonomously and possess a common objective and concern.
Farmers Producer Companies	: The FPC, or Financial Participation Company, represents a synthesis of cooperative groups and private limited companies. The Farmer Producer Companies, which are officially incorporated in accordance with the Indian Companies Act of 2013, exhibit democratic governance wherein every producer or member is granted equal voting rights, regardless of their shareholding.
Gross Domestic Product	: The concept of Gross Domestic Product (GDP) refers to the comprehensive monetary or market assessment of all finalized goods and services generated within the geographical confines of a nation during a designated period.
ICAR	: Indian Council of Agricultural Research (ICAR).
Soil Health Card (SHC)	: A soil health card provides information on the nutrient status of soil , along with recommendations on the dosage of nutrients to be utilised for improving its fertility and health.
Weather Based Crop	: It is intended to provide insurance protection to

16.11 SUGGESTED READINGS/ REFERENCES

1. Agriculture Schemes Paperback -Dr. Sanjeev Atrey (Author).
2. Fundamentals of Agriculture (VOL. 1-2) by R.L. Arya (Author), Sonam Arya(Author), Renu Arya /Janardhan Kumar (Author).
3. Structure and Performance of Agricultural Schemes in India Paperback – Import, 27 June 2012 by KomolSingha (Author).
4. Abrol, I.P., Sangar, S. (2006). Sustaining Indian agriculture–conservation agriculture the way forward. *Curr. Sci.* 91 (8), 1020–1025.
5. Al-Jawaldeh, A., & Abbass, M. M. (2022). Unhealthy dietary habits and obesity: the major risk factors beyond non-communicable diseases in the eastern Mediterranean region. *Frontiers in Nutrition*, 9, 817808.
6. Auffhammer, M., Ramanathan, V., Vincent, J.R. (2012). Climate change, the monsoon, and rice yield in India. *Clim. Change* 111 (2), 411–424. doi:10.1007/s10584-011-0208-4
7. Bhatt, H., Bhushan, B., & Kumar, N. (2019). IOT: the Current Scenario and Role of Sensors Involved in Smart Agriculture. *International Journal of Recent Technology and Engineering*, 8(4), 12011-12023.
8. Carleton, T.A., 2017. Crop-damaging temperatures increase suicide rates in India. *Proc. Natl. Acad. Sci.* 114 (33), 8746–8751.
9. Christiaensen, L., Demery, L., &Kuhl, J. (2011). The (evolving) role of agriculture in poverty reduction—An empirical perspective. *Journal of Development Economics*, 96(2), 239-254.
10. Cummings, R.W. (2019). RS Paroda: reorienting Indian agriculture: challenges and opportunities.
11. Department of Agriculture, Cooperation & Farmers' Welfare (DAC&FW). Annual Report (2020-21). <https://agricoop.nic.in/en/whatsnew>
12. Dhahri, S., & Omri, A. (2020). Foreign capital towards SDGs 1 & 2—Ending Poverty and hunger: The role of agricultural production. *Structural Change and Economic Dynamics*, 53, 208-221.
13. Dhinakaran, D. D. P., & Kesavan, N. (2020). Exports and imports stagnation in India during COVID-19-A Review. *GIS Business (ISSN: 1430-3663 Vol-15-Issue-4-April-2020)*.
14. e-NAM. 2023. <https://www.enam.gov.in/web/>
15. Fanzo, J. (2019). Healthy and sustainable diets and food systems: the key to achieving Sustainable Development Goal 2. *Food ethics*, 4, 159-174.
16. Guiteras, R. (2009). The Impact of Climate Change on Indian agriculture. Manuscript. Department of Economics, University of Maryland, College Park, Maryland.

17. Gupta, N. (2016). Decline of Cultivators and Growth of Agricultural Labourers in India from 2001 to 2011. *Int. J. Rural Manag.* 12 (2), 179–198. doi:10.1177/0973005216665939.
18. Gupta, G. & Nagar, M. (2017). Agriculture Sector in India: As a Career. *International Journal on Arts, Management and Humanities* 6(2): 01-06.
19. INCCA. (2010). India: Greenhouse gas emissions 2007.
20. India Brand Equity Foundation (IBEF) (2021). Indian agriculture and allied industries industry report. <https://www.ibef.org/industry/agriculture-india.aspx>
21. McMichael, A. J. (2001). Impact of climatic and other environmental changes on food production and population health in the coming decades. *Proceedings of the Nutrition Society*, 60(2), 195-201.
22. Ministry of Agriculture and Farmers Welfare. 2023.
23. Raj, S., Roodbar, S., Brinkley, C., & Wolfe, D. W. (2022). Food Security and climate change: Differences in impacts and adaptation strategies for rural communities in the Global South and North. *Frontiers in Sustainable Food Systems*, 5.
24. Rao, C.R., Raju, B.M.K., Rao, A.S., Rao, K.V., Rao, V.U.M., Ramachandran, K., Venkateswarlu, B., Sikka, A.K., Rao, M.S., Maheswari, M., Rao, C.S. (2016). A district level assessment of the vulnerability of Indian agriculture to climate change. *Curr. Sci.* 1939–1946.
25. Ray, L.K., Goel, N.K., Arora, M. (2019). Trend analysis and change point detection of temperature over parts of India. *Theor. Appl. Climatol.* 138 (1–2), 153–167. doi:10.1007/s00704-019-02819-7.
26. Saha, A., Ghosh, S., Sahana, A.S., Rao, E.P. (2014). Failure of CMIP5 climate models in simulating post-1950 decreasing trend of Indian monsoon. *Geophys. Res. Lett.* 41, 7323–7330. doi:10.1002/2014GL061573.
27. Singh, C., Rio, C.R.D., Soundarajan, V., Nath, Nath., Shivaranjani, V. (2019). Assessing India's mounting climate losses to Financial Institutions, <http://www.indiaenvironmentportal.org.in>, accessed on 28th February 2020
28. Soubry, B., Sherren, K., Thornton, T.F. (2020). Are we taking farmers seriously? A review of the literature on farmer perceptions and climate
29. World Bank. 2023. <https://www.worldbank.org/en/topic/agriculture/overview#:~:text=Growth%20in%20the%20agriculture%20sector,more%20than%2025%25%20of%20GDP.>
30. Zaveri, E., Grogan, D.S., Fisher-Vanden, K., Frolking, S., Lammers, R.B., Wrenn, D.H., Prusevich, A., Nicholas, R.E. (2016). Invisible water, visible impact: groundwater use and Indian agriculture under climate change. *Environ. Res. Lett.* 11 (8), 084005. doi:10.1088/1748-9326/11/8/084005.

16.12 ANSWERS TO CHECK YOUR PROGRESS EXERCISES

Check Your Progress Exercise 1

- 1) Agriculture serves as the primary occupation for approximately 58 per cent of India's population. The Gross Value Added (GVA) generated by the sectors of farming, forestry, and fisheries in the fiscal year 2019 (provisional estimate) is around Rs 18.55 lakh crore (equivalent to US\$ 265.51 billion). The agricultural sector plays a crucial role in facilitating socio-economic development. Agriculture serves as the predominant means of sustenance for around 58 per cent of India's population (IBEF, 2021), with almost 70 per cent of rural households relying only on agricultural activities.
- 2) The primary objective of this initiative is to offer insurance coverage and financial assistance to agricultural practitioners in the event of crop failure caused by natural disasters, pests, and diseases. The objective is to establish measures that will enhance the financial stability of farmers, thus facilitating their long-term engagement in agricultural activities.
- 3) The primary objective of the central government is to optimize agricultural production and foster job creation within the agricultural industry. To accomplish this objective, the government of India has implemented a range of governmental initiatives targeting farmers. The primary objective of the central government is to foster the creation of novel services that have the potential to enhance the employment rate and improve agricultural productivity.

Check Your Progress Exercise 2

- 1) A total of 29 Centrally Sponsored Government programmes have been implemented to provide impetus to the agriculture business and augment the income of farmers nationwide. Below are several significant government schemes pertaining to the agricultural sector.

The Soil Health Card Scheme was initiated in the year 2015. The implementation of soil health cards would facilitate farmers in obtaining pertinent information regarding nutrient levels, as well as receiving recommendations for enhancing nutrient content and promoting soil fertility.

The Pradhan Mantri Fasal Bima Yojana is a scheme that operates on an actuarial premium basis. The implementation of Neem Coated Urea serves as a regulatory measure for the utilization of urea. The Pradhan Mantri Krishi Sinchai Yojana is a government initiative aimed at addressing the irrigation needs of the agricultural sector. The Parampragat Krishi Vikas Yojana is an initiative aimed at fostering the practice of organic farming.

- 2) Since agriculture is a state responsibility, the above plans and initiatives must be implemented in collaboration with the State Governments. The

government supports these initiatives by increasing budgetary allocation and providing non-budgetary financial resources, such as Corpus Funds like the Micro Irrigation Fund, Agriculture Infrastructure Funds, PM Matasya Sampada Yojana, Animal Husbandry Infrastructure Development Fund, and Farmer Producer Organizations (FPOs) and Gramin Agriculture Markets. These programs have also increased foodgrain and horticulture crop production. The nation's agricultural exports have also increased. The agriculture sector's budget increased from Rs. 27662.67 crore in 2013-14 to Rs. 1,32,513.62 crore in 2022-23, enabling these achievements. The latest Economic Survey shows that agricultural and allied industries grew 3.6 per cent in 2020-21.

- 3) Farmers need to be educated about agricultural literacy and the benefits of various initiatives. Government-implemented agricultural projects should follow strict procedures to maximize benefits for beneficiaries. Relax regulations and protocols for government-issued agricultural credit to increase accessibility. Additionally, these loan facilities should have lower interest rates.

